

Investigation & Evidence

Use the cue, then fade the cue. Study the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS A fair test follows a cycle: **ask a question**, **predict** what will happen, **plan a fair test**, **observe & measure**, **record data**, and **explain from evidence**. In a fair test you change **one thing** and keep everything else the same. A conclusion must be backed up by the data you collected.

Round 1 — Retrieve It

1. What is the first step you take when you start an investigation?

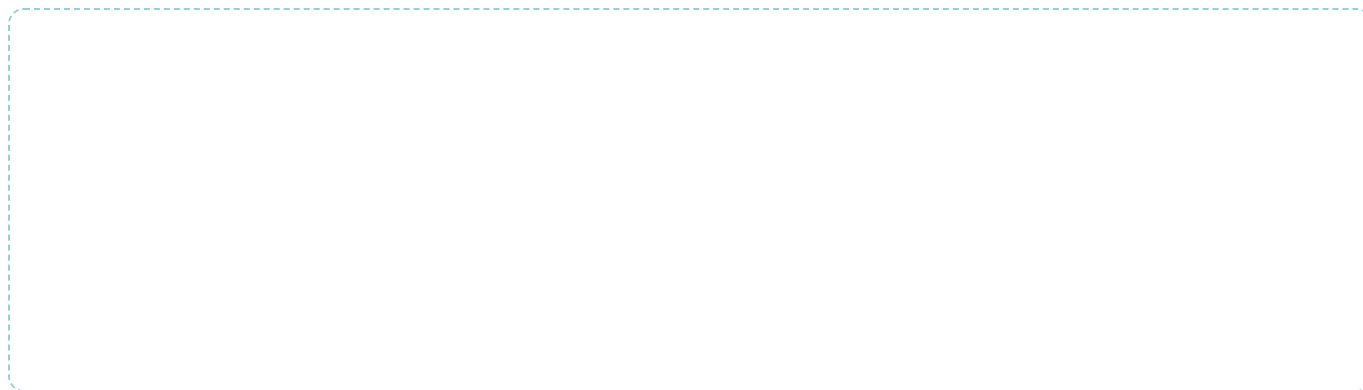
2. In a fair test, how many things should you change at a time?

3. What do we call the careful facts and numbers you write down while you observe?

4. A good conclusion must be supported by what?

Round 2 — Sketch It

Draw the cycle of a fair test as a loop with arrows. Label each step in order: ask a question, predict, plan a fair test, observe & measure, record data, and explain from evidence.



Investigation & Evidence — Apply It

Keep the cue card covered. Answer first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Cycle

1. You want to find out if plants grow taller with more sunlight. Write a question and a prediction for this test.

2. To test the plants fairly, name the one thing you would change and two things you would keep the same.

3. What tool and unit would you use to measure and record how tall each plant grows?

Round 4 — Reason from Evidence

A student tested which paper towel soaks up more water. One towel was dipped for 5 seconds and the other for 20 seconds. The student says the longer towel is "the best." Is this a fair test? Explain what went wrong and how to fix it.

Investigation & Evidence — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain why a fair test changes only one thing at a time. What would happen to your conclusion if you changed two things at once?

C**Connect**

Connect "record data" to "explain from evidence." Why can't you trust a conclusion if you never wrote down your observations?

E**Extend**

Design your own fair test for a question you are curious about. List your question, the one thing you will change, what you will keep the same, and what data you will record.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Investigation & Evidence

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Ask a question (about something you notice or wonder).
2. Only one thing (one variable); keep everything else the same.
3. Data (observations and measurements).
4. The evidence (the data you collected during the investigation).

Sketch It — look for

A loop with arrows showing all six steps in order: ask a question → predict → plan a fair test → observe & measure → record data → explain from evidence.

Apply It

1. Question example: "Do plants grow taller with more sunlight?" Prediction example: "I think plants in more sunlight will grow taller."
2. Change: the amount of sunlight. Keep the same (any two): same plant type, same pot and soil, same amount of water, same start height.
3. A ruler (or meter stick), measured in centimeters (cm), recorded in a table.

Reason from Evidence — look for

Not a fair test. Two things changed (the towel and the dip time), so we cannot tell which caused the difference. Fix it by dipping both towels for the same amount of time and changing only the towel.

ACE — Extend look-for

Plan changes one variable, holds the others constant, and names data that answers the question (question → one change → controlled variables → data to record).

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color.